

Anticoagulant & Antiplatelet:

Dr kabsha course

00201064018621



Anticoagula

→ Indirect thrombin Inhibitors

→ Direct thrombin inhibitors

→ Direct factor Xa inhibitors

→ Vit K antagonist

❖ Indirect thrombin Inhibitors

(Heparin)

enhance antithrombin against several serine proteases of the coagulation system, most importantly factors IIa (thrombin)

★ UFH (mucopolysaccharide polymer):

Shorter t_{1/2}, more to cause Heparin inducing thrombocytopenia (HIT) ttt by argatroban.

✚ Monitor aPTT, Antidote is protamine sulfate

✚ Heparin use in Dialysis patient (if crcl <15 ml/min)

★ LMWH (enoxaparine):

Dose 1 mg/kg/dose BID, 1.5 mg/kg/ dose OD

✚ Pregnant we use LMWH, but if she going to labor we use UFH due to shorter ½

Enoxaparin (CLEXAN) dose

▲ As Treatment VTE and NSTEMI

The general dose

1mg/kg/dose **BID**

OR 1.5mg/kg/dose **OD**

▲ Treatment STEMI

Adult like above.

Age > 75 year

0.75mg/kg/dose **BID**

▲ As prophylaxis

In general 40mg >> 4000 I.U **OD**

▲ In obesity patient

• BMI (BODY MAX INDEX) < 50 kg/m² dose ; 40mg **BID**

• BMI > 50 kg/m²; 60 mg **BID**

▲ In renal impediments

Need dose adjust if CrCl < 30 mL/minute:

• As prophylaxis 30mg **OD**

• As treatment 1mg /kg/dose **OD**

▲ In children

• As prophylaxis ;

< 2 month 0.75mg/kg/ dose **BID**

≥ 2 month 1mg/kg/dose **BID**

• As treatment

1 to < 2 months:: 1.5 mg/kg/dose **BID**

≥ 2 months: 1 mg/kg/dose **BID**

• NOTE;

IF CrCL < 15ml/minute convert patient to HEPARIN

TO CONVERT DOSE FROM MG TO I.U USE THIS CALCULATION

1mg/kg*100

EXAMPLE : Age 60 year calculate the Dose of enoxaparin as treatment

VTE.

1*60*100 = 6000 u **BID**.

❖ **Direct thrombin inhibitors:**

★ **Oral**

Dabigatran# GI side effect (dyspepsia), antidote idarucizumab

Dapigatran t1/2 12-18 hrs,,,, C.I in renal failure

★ **Parenteral**

Bivalirudin, lepirudin, desirudin, Argatroban# use in case of Heparin inducing thrombocytopenia (HIT)

❖ Direct factor Xa inhibitors

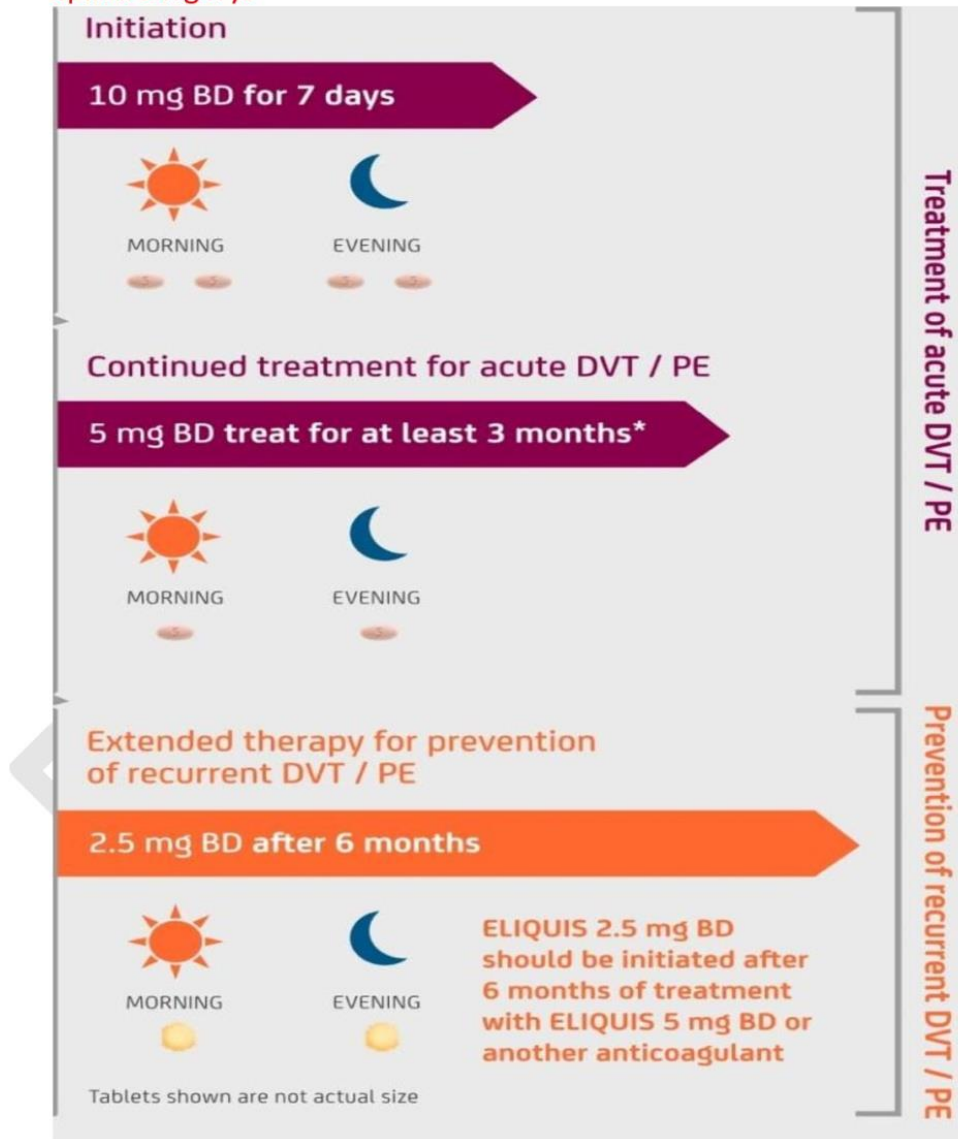
★ Oral

Rivaroxaban, Apixaban..... antidote ~~~andexanet (xa)

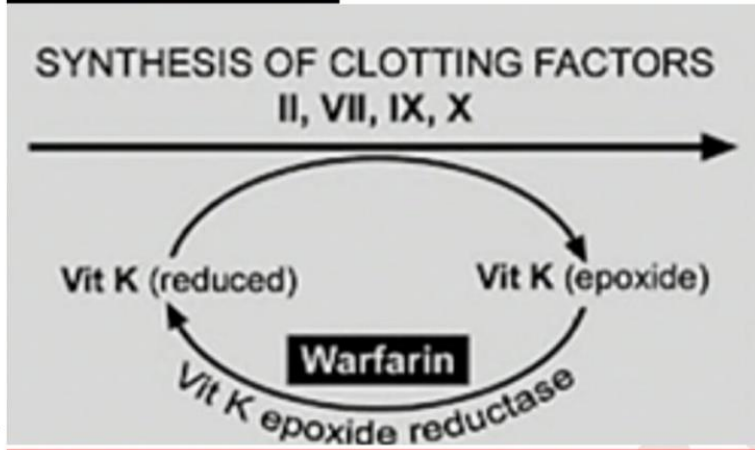
★ Parenteral

Fondaparinux (pentasaccharide) structure??

✚ Direct factor Xa inhibitors **NOT** use in sitting lumbar puncture (LP) or spinal surgery.



❖ Vit K antagonist



(Warfarin)

- ✚ **SE:** Hemorrhage, Purple toe syndrome
- ✚ Cross BBB, Category **X** à cause Nasal bone hypoplasia
- ✚ **Avoid:** Tamoxifen, SERM/estrogen à increase bleeding
- ✚ .monitor by **INR**
- ✚ **Antidote:** Vit K
- ✚ Vit **b12** can be taking with warfarin, **warfarin with AL** complex in GI & poorly absorption

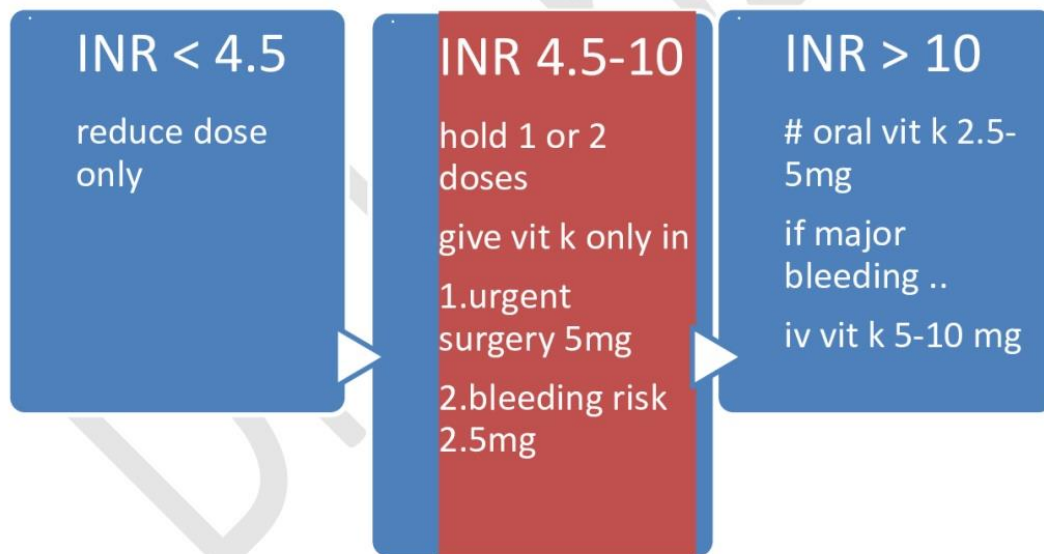
A.fib + mechanical valve: warfarin, dabigatran, Rivaroxaban

Valve replacement: warfarin

Anticoagulant take with meal ☐: Rivaroxaban

Normal range of **INR** on warfarin therapy → **2-3**

In mechanical mitral valve → **2.5-3.5**



7 A patient is taking warfarin 4.5 mg PO daily came to the clinic with INR of 5.1 without any signs of active bleeding, what is the appropriate management for her?

- A. Hold warfarin and give vit.K IV
- B. Hold warfarin and resume with lower dose
- C. Hold warfarin and give platelet

Herbal interact with warfarin **decrease** INR S C G E Vit k

S...soya, st.jhon warts

C... co enzyme Q??

G... Green tea

E...echinachea

ginseng?

Enzyme inhibitor & inducer

cyp --

AB, antifungal, omeprazole

Grape fruit, alcohol??

Valporoic acid

cyp +

rifampicin

GPS.....grisofulvin...p....sulphonyl

st.jhon wart

carbamazepine..phenytion

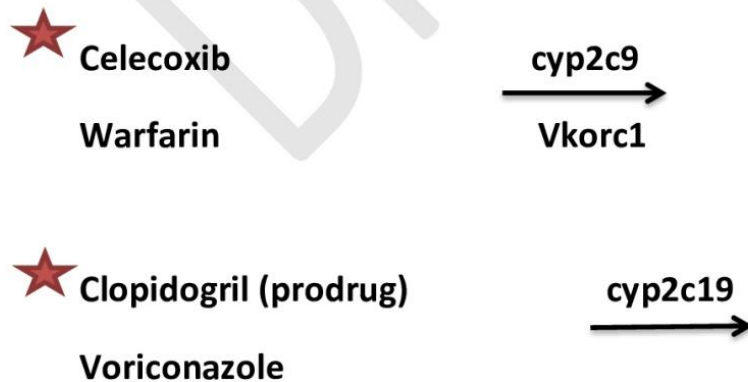
Smoking

Amiodarone and warfarin drug interaction:

 **Increase warfarin effect**

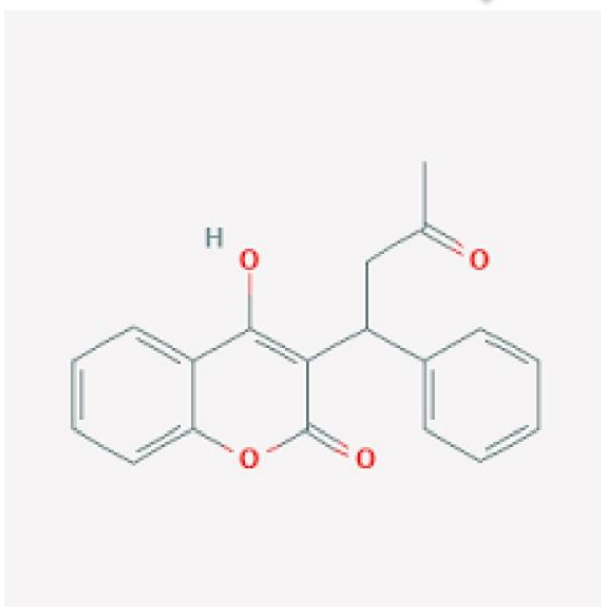
Warfarin – Pharmacokinetic Drug Interactions

- Warfarin is a substrate of CYP 2C9 (major), 1A2 (minor), 2C19 (minor) and 3A4 (minor) and an inhibitor of 2C9 (weak) and 2C19 (weak). Avoid use with tamoxifen.
- 2C9 inducers can ↓ INR; these include aprepitant, bosentan, carbamazepine, phenobarbital, phenytoin, primidone, rifampin (large ↓ INR), licorice and St. John's Wort.
- 2C9 inhibitors can ↑ INR; these include amiodarone, azole antifungals (e.g., fluconazole, ketoconazole, voriconazole), capecitabine, etravirine, fluvastatin, fluvoxamine, macrolide antibiotics, metronidazole, tigecycline, TMP/SMX and zafirlukast. See Drug Interactions chapter.
- When starting amiodarone, ↓ the dose of warfarin by 30 – 50%.



Structure?

Phase II conjugation



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❖ ANTIPLATELET :

- ➡ Inhibit prostaglandin synthesis
- ➡ P12Y12 receptor inhibitors
- ➡ Glycoprotein IIb /IIIa inhibitor
- ➡ PAR receptor inhibitor.

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❖ Inhibit prostaglandin synthesis

(Aspirin) >>> ibuprofen??

- ✚ Inhibits synthesis of **thromboxane A2** by **irreversibly** acetylation of COX enzyme
- ✚ **NOT** give in children and teenagers (< 20 years) à **Reyes syndrome** "tinnitus of ear"
- ✚ **Not** use in **asthmatic** patient, and it is **CI** with **methotrexate**

❖ Pa2Y12 receptor inhibitors

(Clopidogrel, Ticlopidine, Ticagrelor, prasugrel)

- ✚ Clopidogrel is **CYP 2C19**, and can **NOT** take it with omeprazole, use Pantoprazole

❖ glycoprotein IIb /IIIa inhibitor

- ✚ **abciximab** ,,, **epifibatide** ,, **tirofiban**

❖ PAR receptor inhibitor

- ✚ **Varpaxar** ,,, **atopaxar**
- ✚ **Dipyridamole** >>>> **V.D & anti platelet**

- Fibrinolytic drug : (destroy formed clot)

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Plasmenogen → plasmin

Streptokinase ..urokinase // alteplase

IMPORTANT NOTES

❖ STOP BEFORE SURGERY

Enoxaparin → before 1 day

Warfarine → before 5 days

Clopidogril → before 5 days

UFH → before 4-6 hr.

LMWH → before 24 hr.

INR → warfarin

APTT → heparin

Heparin → renal patient

Urgent Cardiac surgery of PCI required: Bivalirudin

→ **Score: CHADS2 vs CHA2DS2 VASC**

CHADS2 Atrial fibrillation

CHF...1 Age≥75years...1

HTN..1 DM ...1 STROKE2

RESULT...

ZERO... no ttt

1..... Antiplatelet(aspirin,clopidogril)

2..... Anticoagulant (dapigatran)

Exam ... score 5 or calculate score

CHA2DS₂ VASC

Age ≥ 75 2

65-74 yr... 1

(female)...1

Vasc (prior MI/PAD/ aortic plaque)....1

Result

Male 0 or female 1 ...no ttt

Male 1 or female 2 Dabigatran

Oral anticouglant is recomnded (non vitamin k oral DOAC) OVER warfarin

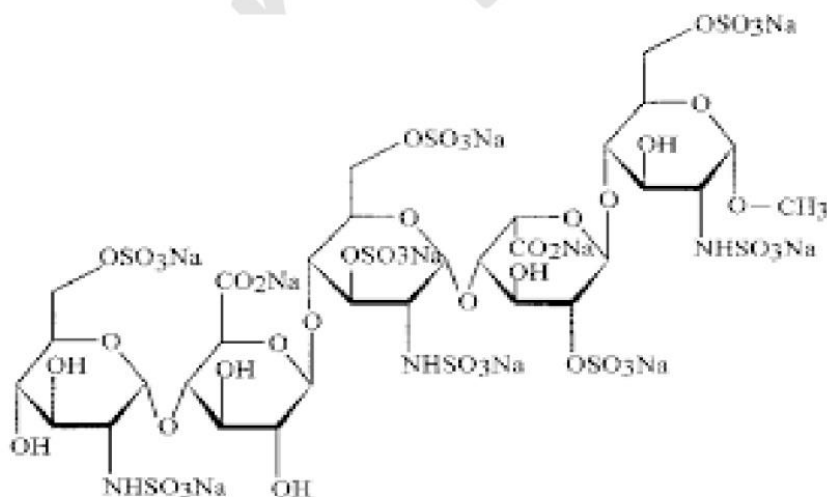
Which of the following is the recommended anticoagulation strategy for atrial fibrillation patient with a CHA2DS2-VASc score of 5?

- A. aspirin
- B. dabigatran**
- C. clopidogrel
- D. no anticoagulation

80- A 76-year-old cardiac male patient was transferred to the hospital for palpitation with a heart rate of 177 bpm. He is a heavy smoker and his past medical history includes hypertension, hyperlipidemia and coronary artery disease. His medications include aspirin, clopidogrel, carvedilol, valsartan and atorvastatin. He was diagnosed as having atrial fibrillation.

Which of the following is the CHA2DS2-VASc score for this patient?

- ☐ A. 1
- ☐ B. 3
- ☒ C. 4
- ☐ D. 6



- ❖ Anti platelet cause dyspnea : ticagrelor
- ❖ Anti platelet C.I in TIA: prasugrel
- ❖ Anti couglant cause dyspepsia: dabigatran
Bridge?)

Notes:

Treatment Provoked.. → 3 months
Nonprovoked ... → 6months

DVT **prophylaxis** in postpartum .. → 6weeks

Administering one intramuscular (IM) dose of vitamin K

.... routinely to all newborns within the first 6 hours post-birth

But in coagulopathy taken iv

Drug ttt bleeding :

Aminocaproic acid .. tranxemic acid ...

<https://www.facebook.com/groups/477348776664047/?ref=share>

-UFH Cause

- a)Hypokalemia**
- b)Hyperkalemia**
- c)Hypocalcemia**
- d)Hypercalcemia**

-which of the following parenteral anticoagulants require routine monitoring of coagulation lab parameters?

- A) UFH intravenously**
- B)UFH subcutaneously**
- C)enoxaparin subcutaneously**
- D)fondaparinux subcutaneously**

-Which of the following is the preferred anticoagulant therapy during hemodialysis procedure?

- A. Enoxaparin**
- B. Dabigatran**
- C. Rivaroxaban**
- D. Unfractionated heparin(UFH)**